



How data analysts approach tasks

The data analysis process is composed of the six phases of **ask**, **prepare**, **process**, **analyze**, **share**, and **act**. Their purpose is to gain insights that drive informed decision-making. Earlier in this course, you explored a case study about a group of data analysts using the six phases of data analysis to improve employee retention at their company. In this reading, you'll focus on the phases themselves and explore how a data analyst might use this process to help a fictional company make data-driven decisions about investing in training.



The six phases of data analysis

The data analysis process helps analysts break down business problems into a series of manageable tasks:

In the **ask phase**, you'll work to understand the challenge to be solved or the question to be answered. It will likely be assigned to you by stakeholders. As this is the ask phase, you'll ask many questions to help you along the way.

Next, in the **prepare phase**, you'll find and collect the data you'll need to answer your questions. You'll identify data sources, gather data, and verify that it is accurate and useful for answering your questions.

The **process phase** is when you will clean and organize your data. Tasks you perform here include removing any inconsistencies; filling in missing values; and, in many cases, changing the data to a format that's easier to work with. Essentially, you're ensuring the data is ready before you begin analysis.

The **analyze phase** is when you do the necessary data analysis to uncover answers and solutions. Depending on the situation and the data, this could involve tasks such as calculating averages or counting items in categories so you can examine trends and patterns.

Next comes the **share phase**, when you present your findings to decision-makers through a report, presentation, or data visualizations. As part of the share phase, you decide which medium you want to use to share your findings and select the data to include. Tools for presenting data visually include charts made in Google Sheets, Tableau, and R.

Last is the **act phase**, in which you and others in the company put the data insights into action. This could mean implementing a new business strategy, making changes to a website, or any other action that solves the initial problem.

Putting the process into practice

Now, think about how the phases in this process can be applied to a business situation.

The retirement contribution dilemma

The management team at a fictional midsize tech company, Geo-Flow, Inc., noticed that employee participation in the company's retirement contribution program was lower than expected. The company had invested a lot of resources in establishing its world-class benefit program, with the goal of reducing employee turnover. Because so few employees were using the program, leaders wondered if they should develop educational training to explain the benefits to employees. They wanted to make a well-informed decision before committing to the investment, so they asked their data analytics department to make a recommendation.

The analysts used the six phases framework and began by defining the problem. They **asked**, "Are employees investing in the company's retirement contribution program?" And, if not, "Should we create an educational program to encourage participation?" Satisfied with their research questions, they **prepared** their analysis project by gathering data from HR, such as employee demographics, salary levels, and current retirement contributions.

Next, they **processed** the data by cleaning and organizing it. They removed duplicates and data from individuals who had retired or left the company, then sorted the data by the employees' ages, departments, and length of employment. Their **analysis** showed that some employee groups were less likely to contribute to the plan or to be aware that the company offers a matching contribution. They interpreted these results to mean that these employee groups were not receiving enough education on the company's retirement contribution matching program. They also studied the data to find trends and insights and used data visualization to review their analysis by exploring it in different contexts.

The analysts **shared** their findings with the management team using visualizations including bar and pie charts that illustrated the facts clearly so decision-makers could easily interpret the data. The report showed that, while overall participation was decent, some employee groups were not taking full advantage of the retirement program—but they might, if they knew more about the program and the matching contribution the company offers.

Based on these findings, the company took **action**, creating a targeted educational program focusing on the benefits of retirement contributions, specifically aimed at the employee groups identified as low contributors. Results showed that a few months after implementing this training, there was a significant increase in retirement contributions among the targeted groups.

Iteration during the data analysis process

The data analysis process is designed to build on itself, so the results from each step are the inputs for the next step. Keep in mind, however, that you might not always move through the steps linearly. For example, you might be in the analyze phase and find out your data was pulled from the wrong database. Or, you could learn while cleaning the data that your original question didn't adequately define the problem.

In cases such as these, you may have to go back to an earlier stage and work through the process with new, better information. The important thing is not to skip steps and miss something that's important. In fact, the biggest mistake analysts make when using this framework is looking for quick and easy answers.

Finally, **make sure to review your work in each phase of the analysis.** This helps you learn more about the situation and your own skill set, which will lead to the kind of continuous growth that helps data professionals succeed.

Key takeaways

The six phases of the data analysis process help answer business challenges, such as understanding how to improve a retirement program. Additionally, iterating on and reviewing your work throughout the data analysis process is critical for obtaining quality results.